

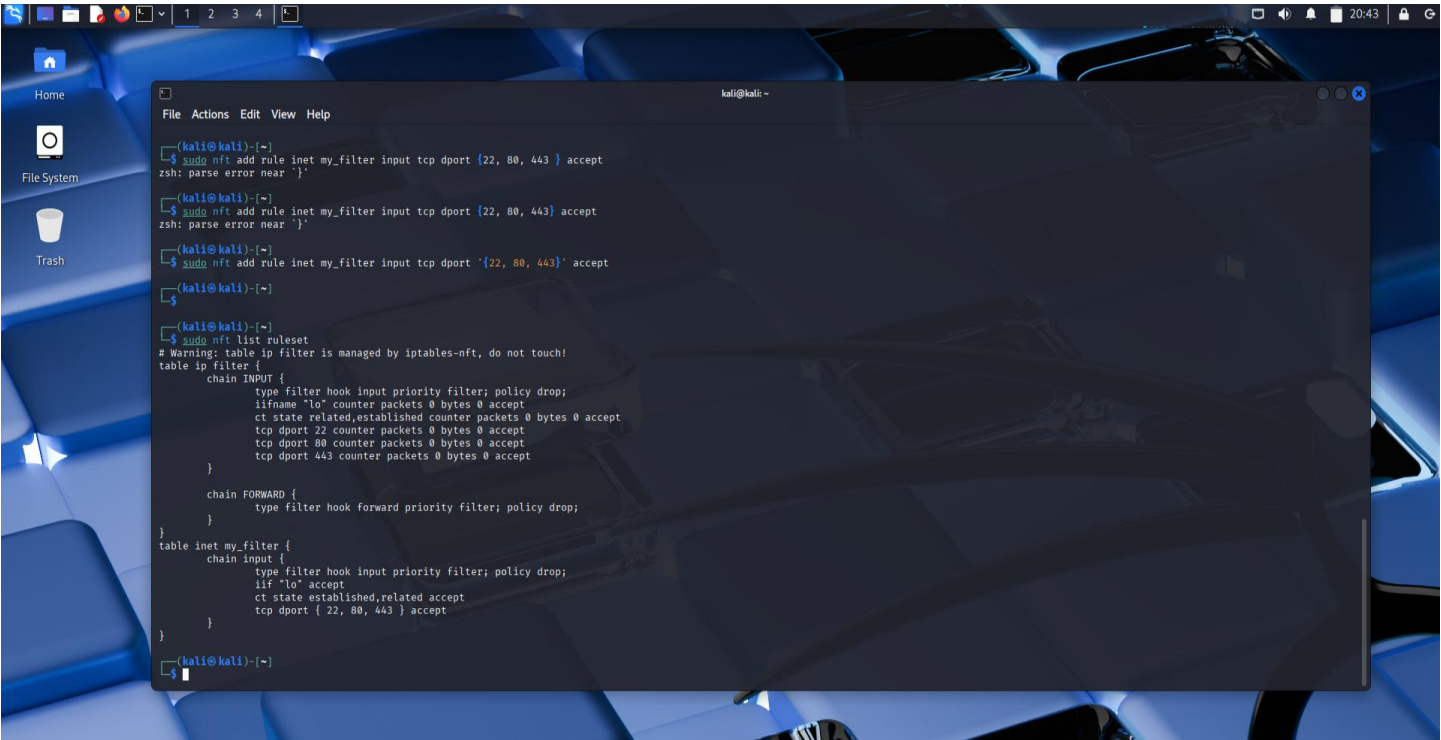
INFOE 4

Lab Activity: Foundations of Linux Firewalls (iptables & nftables)

Screenshot of iptables-L-n-v showing all 5 rules (Loopback, State, SSH, HTTP, HTTPS).



Screenshot of nft list ruleset.



Critical Thinking Questions (Submission)

1. Stateful vs. Stateless: Why did we need the ESTABLISHED, RELATED rule? What would happen to your sudo apt update command if that rule were missing, even if OUTPUT is set to ACCEPT?

- **Answer:** We need the ESTABLISHED, RELATED rule because a stateful firewall remembers the connections that you initiated. If this rule were missing, your sudo apt update command would fail. Even though your server is allowed to send the request out (because OUTPUT is ACCEPT), the update server's reply would be blocked from coming back in because your default INPUT policy is to DROP everything. The stateful rule acts as a VIP pass, allowing return traffic for sessions you started.

2. Order of Operations: If you placed a DROP ALL rule at the very top of your INPUT chain, would your SSH rule at the bottom still work? Explain why.

- **Answer:** No, the SSH rule would not work. This is because iptables uses a "First Match Wins" logic. It reads your rules from top to bottom. If the very first rule it reads tells it to drop everything, the firewall applies that action immediately and stops evaluating the rest of the chain. It would drop your SSH traffic before ever seeing the allow rule at the bottom.

3. nftables Advantage: Identify one syntax difference between iptables and nftables that makes the latter easier to manage for large-scale networks.

- **Answer:** One major advantage is that nftables allows you to combine multiple ports into a single, readable string using curly braces. For example, you can allow SSH, HTTP, and HTTPS all at once with tcp dport {22, 80, 443 } accept. In iptables, you had to write a completely separate, repetitive rule for each individual port. This makes nftables much cleaner and easier to manage when dealing with large amounts of rules.

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